

# CAFE SALES ANALYSIS TECHNICAL REPORT

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Data Cleaning, Exploratory Analysis & Dashboard Insights  
Financial Year 2023

|         |                                                  |
|---------|--------------------------------------------------|
| Dataset | Cafe Sales (dirty_cafe_sales → cafe_sales_clean) |
| Period  | January 2023 – December 2023                     |
| Tools   | MySQL · Power BI Desktop                         |
| Records | 9,006 raw → 8,984 clean (22 removed)             |

## 1. Executive Summary

This report presents a full data analysis of cafe sales transactions recorded during the 2023 financial year. The analysis covers data cleaning performed in MySQL, exploratory SQL queries, and an interactive Power BI dashboard built on the cleaned dataset.

The cafe generated total revenue of R76,518 across 8,576 transactions in 2023, with an average order value of R8.92. Salad was the single best-performing item, contributing to R14,890 (19.5%) of total revenue. Revenue and transaction volumes were broadly consistent from month-to-month, with no extreme seasonal peaks. Payment methods were almost evenly split among Cash, Credit Card, and Digital Wallet for attributed records. In-store and Takeaway channels were also near-identical in both volume and revenue.

A significant data quality limitation affects this analysis: approximately 31% of all records carry UNKNOWN values in the location and/or payment method fields. These records are retained and disclosed throughout but limit the depth of channel and payment-method conclusions.

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## 2. Data Cleaning

### 2.1 Process Overview

The raw dataset (`dirty_cafe_sales`) contained 9,006 rows with non-standard column names, mixed-format values, placeholder strings (UNKNOWN, ERROR), and NULL entries across several fields. Cleaning was performed in MySQL using a multi-step approach: column renaming, staging table creation, duplicate detection, row deletion, value imputation, and constraint enforcement.

### 2.2 Step-by-Step Summary

#### Step 1 – Column Renaming

All eight columns were renamed from title-case, space-separated names to snake\_case (e.g. Transaction ID' → `transaction_id`) to ensure compatibility with SQL operations and Power BI field naming conventions.

#### Step 2 – Staging Table

A staging table (`cafe_sales_staging`) was created as an exact copy of the renamed source table to preserve the original data. All subsequent transformations were applied to the staging copy, and the final clean table (`cafe_sales_clean`) was derived from it.

#### Step 3 – Duplicate Removal

A CTE using `ROW_NUMBER()` partitioned over all eight columns was used to identify exact duplicate rows. Only the first occurrence of each unique row was carried forward into `cafe_sales_clean`.

#### Step 4 – Row Deletion (Multi-field Null Records)

Rows where items, payment\_method, and locations were all simultaneously missing or blank were deleted entirely, as these records contained no recoverable analytical value. This accounted for the majority of the 22 rows removed.

### Step 5 – total\_spent Imputation

Records with NULL, blank, ERROR, or UNKNOWN values in the total\_spent column were recalculated using the formula:  $\text{total\_spent} = \text{quantity} \times \text{price\_per\_unit}$ . No records had missing quantity or price\_per\_unit values, so imputation was successful for all affected rows.

### Step 6 – payment\_method & location Standardisation

Rather than deleting records with missing payment\_method or location values, these were standardised to the literal string 'UNKNOWN'. This preserves the records for revenue and transaction counting while clearly flagging them as unattributed in any segmentation analysis.

### Step 7 – transaction\_date Standardisation

Date values that were NULL, blank, ERROR, or UNKNOWN were set to NULL. This ensures date-based queries and time-series visuals correctly exclude these records rather than erroring or producing misleading results.

### Step 8 – Type Enforcement

Final ALTER TABLE statements enforced correct data types: total\_spent as DOUBLE and transaction\_date as DATE, ensuring downstream compatibility with Power BI's data model and DAX calculations.

## 2.3 Column-Level Cleaning Reference

| Column           | Data Type | Issues Found                   | Resolution                                                       |
|------------------|-----------|--------------------------------|------------------------------------------------------------------|
| transaction_id   | VARCHAR   | None                           | None                                                             |
| item             | VARCHAR   | Blanks / UNKNOWN / ERROR       | Rows where item + payment + location all missing → deleted       |
| quantity         | INT       | None found                     | None                                                             |
| price_per_unit   | DECIMAL   | None found                     | None                                                             |
| total_spent      | DECIMAL   | NULL / blank / ERROR / UNKNOWN | Recalculated as $\text{quantity} \times \text{price\_per\_unit}$ |
| payment_method   | VARCHAR   | NULL / blank / ERROR / UNKNOWN | Standardised to 'UNKNOWN'                                        |
| location         | VARCHAR   | NULL / blank / ERROR / UNKNOWN | Standardised to 'UNKNOWN'                                        |
| transaction_date | DATE      | NULL / blank / ERROR / UNKNOWN | Set to NULL; excluded from time analysis                         |

## 2.4 Record Count Reconciliation

**Raw records (dirty\_cafe\_sales):** 9,006

**Clean records (cafe\_sales\_clean):** 8,984

**Records removed:** 22 (multi-field null rows)

**Retention rate:** 99.8%

### 3. Exploratory Data Analysis

#### 3.1 Overall Performance

The following headline metrics were derived from the cleaned dataset and validated against the Power BI data model.

| Metric                       | Value                            |
|------------------------------|----------------------------------|
| Total Revenue                | R76,518                          |
| Total Transactions           | 8,576                            |
| Total Quantity Sold          | 27,167 units                     |
| Average Order Value (AOV)    | R8.92                            |
| Best Selling Item (Revenue)  | Salad – R14,890 (19.5% of total) |
| Best Selling Item (Quantity) | Coffee – 3,212 units             |

#### 3.2 Revenue and Transactions by Item

All eight menu items had broadly similar transaction counts (918–1,020), indicating consistent menu appeal across the range. Revenue differences are therefore driven primarily by price per unit rather than popularity. Salad (R15.10 avg), Smoothie (R12.21), and Sandwich (R12.04) are the premium items by unit price; Cookie (R2.99) and Tea (R4.48) are the lowest-priced items and, correspondingly, the lowest revenue contributors.

Coffee is notable: it is the top-selling item by quantity (3,212 units) and transactions (1,012) yet ranks fifth by revenue due to its low unit price (R6.13). This highlights an upselling or bundling opportunity – small increases in Coffee pricing or pairing promotions with higher-margin items could have a meaningful revenue impact.

| Item     | Total Revenue | Transactions | Qty Sold | Avg Price |
|----------|---------------|--------------|----------|-----------|
| Salad    | R14,890       | 985          | 3,120    | R15.10    |
| Sandwich | R11,680       | 969          | 3,074    | R12.04    |
| Smoothie | R11,608       | 951          | 3,033    | R12.21    |
| Juice    | R9,108        | 1,020        | 3,187    | R8.93     |
| Cake     | R9,009        | 989          | 3,180    | R9.10     |
| Coffee   | R6,210        | 1,012        | 3,212    | R6.13     |
| Cookie   | R2,775        | 929          | 2,898    | R2.99     |
| Tea      | R4,179        | 918          | 2,954    | R4.48     |

#### 3.3 Payment Method Analysis

Among attributed transactions, Cash, Digital Wallet, and Credit Card are almost perfectly balanced in both revenue and transaction count – each contributing approximately R17,550–R17,740 (33%) of attributed revenue. Average order values are also near-identical across all three methods (R8.93–R9.07), suggesting that payment method does not influence spending behaviour.

The 2,684 UNKNOWN payment records account for R23,518 – more than any single known payment method. Resolving these records in the source system would significantly improve the analytical value of payment channel reporting.

| Payment Method | Revenue    | Transactions | Avg Order Value |
|----------------|------------|--------------|-----------------|
| Cash           | R17,738.50 | 1,957        | R9.07           |
| Digital Wallet | R17,711.00 | 1,980        | R8.94           |
| Credit Card    | R17,550.50 | 1,955        | R8.98           |
| UNKNOWN        | R23,518.00 | 2,684        | —               |

### 3.4 Location Analysis (In-store vs. Takeaway)

Of the 5,189 attributed transactions, 2,602 (50.1%) were In-store and 2,587 (49.9%) were Takeaway. Revenue was similarly balanced: R23,428.50 (50.7%) In-store vs R22,787.00 (49.3%) Takeaway. This near-perfect split suggests the cafe's two channels operate independently and symmetrically, with no measurable preference between them among customers whose location was recorded.

| Location | Revenue    | Transactions | % of Known Txns |
|----------|------------|--------------|-----------------|
| In-store | R23,428.50 | 2,602        | 43.2%           |
| Takeaway | R22,787.00 | 2,587        | 42.7%           |
| UNKNOWN  | R30,302.50 | 3,387        | —               |

### 3.5 Item Performance by Location

SQL analysis using a RANK() window function (partitioned by location) identified the top item per channel. Salad ranked first for both In-store and Takeaway, reinforcing its position as the cafe's anchor revenue product regardless of channel. This consistency suggests that Salad is a strong candidate for promotional focus, as demand appears channel-agnostic.

### 3.6 Monthly and Daily Trends

Revenue was stable across all twelve months of 2023, ranging from approximately R6,018 (February) to R6,658 (June) – a variance of just 10.6% between the lowest and highest months. There is no evidence of strong seasonal demand. The Power BI dashboard confirms consistent month-over-month performance with no outlier months.

Day-of-week analysis (derived from transaction\_date) shows broadly even daily distributions, with Monday the slightly busiest day by transaction count and Sunday the quietest. The differences are modest, and no single day dominates significantly.

## 4. Data Quality Assessment

### ⚠ Data Quality Note

Approximately 31% of all transactions contain UNKNOWN values in either the location or payment method field (or both). These records have been retained in the dataset and are included in overall revenue and transaction totals. Where breakdowns by location or payment method are presented, UNKNOWN records are shown separately rather than excluded, to preserve data integrity and full revenue accountability.

The most material data quality issue in this dataset is the prevalence of UNKNOWN values in the location and payment\_method columns. In total, 3,387 transactions (39.5% of all 8,576) carry an UNKNOWN location, and 2,684 (31.3%) carry an UNKNOWN payment method. These do not fully overlap – some records are UNKNOWN in one field but not the other.

The decision was taken during cleaning to retain rather than delete these records. This approach preserves the full revenue figure (R76,518) and prevents survivorship bias in transaction counts. However, it means that any analysis segmented by location or payment method must be interpreted with this limitation in mind.

### 4.1 Data Completeness Metrics

**Data Completeness % (both fields known):** 41.8%

**Records with at least one UNKNOWN field:** 58.2%

**Revenue attributable to fully known records:** R32,090 (41.9% of total)

**Revenue from UNKNOWN records:** ~R44,428 (58.1% of total)

### 4.2 Recommendations for Data Collection

- Mandate payment method captures at point-of-sale – this should be a non-nullable system field.
- Enforce location tagging (In-store / Takeaway) at order entry, either via the POS interface or order channel flag.
- Introduce a regular data audit schedule to catch new NULL or placeholder values before they accumulate.
- Consider a retrospective data recovery exercise to classify UNKNOWN records using any available supporting data (e.g. receipt of logs, terminal IDs).

## 5. Power BI Dashboard

### 5.1 Dashboard Overview

The Power BI dashboard is structured across two pages – Main Dashboard and Breakdown – and connects directly to the `cafe_sales_clean` dataset via a semantic model containing two tables: the main transaction table and a calculated Date table. The model includes 27 DAX measures organized into display folders covering Revenue, Transactions, AOV, Location, Payment Method, Day of Week, and Data Quality.

### 5.2 Main Dashboard Page

The main page provides a top-level operational view with the following components:

- **Four KPI cards: Total Revenue (R77K), Total Transactions (9K), Average Order Value (R8.9), and Best-Selling Item (Salad).**
- Each KPI card includes a period-over-period comparison showing the change versus the previous month (e.g. +9.1% revenue vs previous month).
- Total Revenue by Location – clustered bar chart comparing In-store vs. Takeaway revenue.
- Average Order Value by Month Name – line chart showing AOV trends across the year.
- Top 3 Items by Quantity Sold – horizontal bar chart (Coffee, Juice, Cake).
- Total Revenue and Quantity Sold by Month – combo chart with bar (revenue) and line (quantity).
- Total Revenue and Transactions by Month – combo chart showing both volume metrics over time.
- Slicers: Payment Method, Location, Item, and Month Name for cross-filtering.

### 5.3 Breakdown Page

The Breakdown page focuses on data quality transparency and channel/payment segmentation:

- **Three data quality KPI cards: Data Completeness % (41.8%), % Records with Unknown Data (58.2%), Known Records Only Revenue (R32K).**
- Revenue by Payment Method – donut chart showing the three-way split (Cash ~33%, Digital Wallet ~33%, Credit Card ~33%).
- Revenue by Location – bar chart (In-store vs. Takeaway, excluding UNKNOWN).
- Transactions by Day of Week – horizontal bar chart sorted Monday to Sunday.
- Slicers: Year, Month Year, and Day for time and day-level filtering.

### 5.4 Semantic Model Notes

The Date table is a calculated table containing Date, Month Year, Month Number, and Year columns. A Year column was added to support year-level slicing and future multi-year comparisons. All time-intelligence measures (Current Revenue, Previous Revenue, Revenue Change %) use the Date table via DAX `CALCULATE` and `DATEADD` functions. Period-over-period measures are displayed with directional labels and conditional colour formatting (green for positive, red for negative).



## 6. Key Findings & Recommendations

### 6.1 Key Findings

- Total 2023 revenue was R76,518 across 8,576 transactions, with an AOV of R8.92.
- Salad is the top revenue item (R14,890; 19.5%) and the best performer in both In-store and Takeaway channels.
- Coffee is the most popular item by volume (3,212 units) but generates only R6,210 due to its low unit price – a clear upsell opportunity.
- Payment methods and location channels are almost perfectly balanced for attributed records, suggesting no strong customer preference.
- Monthly revenue is stable throughout the year (R6,018–R6,658), with no significant seasonal variation.
- 58.2% of transactions have incomplete location or payment data, materially limiting channel and payment analysis.

### 6.2 Recommendations

- Introduce Salad + Coffee bundle promotion. Salad's high margin and Coffee's high volume make them natural pairing candidates.
- Investigate a modest Coffee price increase (e.g. from R6.13 to R7.00) – at 3,212 units; a R0.87 increase yields ~R2,795 additional annual revenue.
- Prioritize resolution of UNKNOWN data at source (see Section 4.2) to unlock payment and location-level insights.
- Monitor February and September, the two lowest-revenue months, to assess whether targeted promotions could lift performance.
- Explore whether the near-equal In-store / Takeaway split reflects market saturation or untapped growth potential in one channel.

## 7. Appendix – SQL Reference

### 7.1 Data Cleaning Queries (Key Extracts)

The cleaning script performed the following key operations. Full source is available in `cleaning.sql`.

#### Duplicate Detection (CTE with ROW\_NUMBER)

```
WITH duplicates AS ( SELECT *, ROW_NUMBER() OVER ( PARTITION BY transaction_id,
item, quantity, price_per_unit, total_spent,
payment_method, location, transaction_date ) AS dp FROM cafe_sales_staging )
SELECT ... FROM duplicates WHERE dp = 1;
```

#### total\_spent Imputation

```
UPDATE cafe_sales_clean SET total_spent = (quantity * price_per_unit) WHERE
total_spent IS NULL OR total_spent = '' OR total_spent = 'ERROR' OR total_spent =
'UNKNOWN';
```

#### UNKNOWN Standardisation

```
UPDATE cafe_sales_clean SET payment_method = 'UNKNOWN' WHERE payment_method IS NULL OR
payment_method = '' OR payment_method = 'ERROR' OR payment_method = 'UNKNOWN'; --
Same pattern applied to location column
```

### 7.2 Analysis Queries (Key Extracts)

#### Top Item by Revenue per Location

```
WITH revenue_by_item AS ( SELECT location, item, SUM(quantity *
price_per_unit) AS total_revenue FROM cafe_sales_clean WHERE location <> 'UNKNOWN'
AND item <> 'UNKNOWN' GROUP BY location, item ), ranked AS ( SELECT location,
item, total_revenue, RANK() OVER (PARTITION BY location ORDER BY
total_revenue DESC) AS rn FROM revenue_by_item ) SELECT location, item,
total_revenue FROM ranked WHERE rn = 1;
```

#### Busiest Month

```
SELECT YEAR(transaction_date) AS order_year, MONTHNAME(transaction_date) AS
order_month, COUNT(*) AS total_transactions FROM cafe_sales_clean WHERE
transaction_date IS NOT NULL AND item <> 'UNKNOWN' GROUP BY YEAR(transaction_date),
MONTH(transaction_date), MONTHNAME(transaction_date) ORDER BY
total_transactions DESC;
```

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*End of Report*